Adapting DAYS to Golden Gate Assembly for Modular Multigene Pathway Engineering

Research Proposal

Geqi Tie
M.Eng. Candidate in Biological and Biomedical Engineering, Biomanufacturing
McGill University
October 16, 2026

1. Project Summary

This project aims to adapt the Dual-use Assembly Yeast System (DAYS) to a Golden Gate assembly workflow for modular multigene pathway engineering in *Saccharomyces cerevisiae*. The initial focus is to establish a reliable DNA assembly strategy and evaluate high-copy chromosomal integration at delta sites using eGFP as a reporter. The proposed design consists of four plasmids carrying different auxotrophic selection markers, with eight genes involved in β-carotene biosynthesis, precursor supply, lipid metabolism, and redox balance. Assembly accuracy and chromosomal integration will be assessed through genetic screening, sequence verification, fluorescence measurements, and qPCR-based copy-number analysis. Following validation, β-carotene production will be quantified using chromatographic methods.

The overall goal is to establish a modular assembly platform that supports efficient pathway construction, chromosomal integration, and subsequent optimization of metabolite production.

2. Background and Rationale

The Dual-use Assembly Yeast System (DAYS) is a modular platform developed for multigene expression and pathway integration in *S. cerevisiae*. The system uses pDAY vectors containing paired expression units, yeast selection markers, and elements required for donor DNA assembly and chromosomal integration. Unpublished preliminary work in the laboratory has characterized 35 bidirectional promoters and demonstrated CRISPR-Cas9-assisted integration of up to 56 copies of an approximately 8 kb β-carotene pathway at genomic delta sites. These preliminary results provide a foundation for further development of the DAYS platform. The current project focuses on adapting the DNA assembly process to Golden Gate assembly. Unlike conventional restriction-based cloning, Golden Gate assembly uses Type IIS restriction enzymes that cleave outside their recognition sequences. This allows multiple DNA fragments to be assembled in a defined order and orientation using designed overhangs.

In addition, the introduction of standardized Golden Gate assembly interfaces into DAYS could simplify the construction of multigene pathways and allow promoters, coding sequences, and other regulatory elements to be exchanged without redesigning the entire construct. The use of DNA fragments derived from the oligo pool supports parallel assembly of multiple gene combinations. The proposed approach retains the existing DAYS integration strategy while introducing a more modular method for DNA construction and pathway modification.

3. Research Hypothesis and Objectives

Hypothesis

Golden Gate assembly can be incorporated into the existing DAYS platform to enable modular construction of multigene expression cassettes while preserving bidirectional promoter function and chromosomal integration capability.

Research Objectives

Objective 1: Design and validate a Golden Gate-compatible assembly architecture using the existing pDAY vectors.

Objective 2: Establish an eGFP-based screening and quantitative validation workflow to evaluate chromosomal integration and transgene copy number.

Objective 3: Demonstrate the platform using an eight-gene β-carotene production pathway and investigate selected bidirectional promoter configurations for subsequent pathway optimization.

4. Plasmid Design and Pathway Architecture

The proposed system will use four pDAY plasmids, pDU, pDH, pDW, and pDL, carrying the auxotrophic selection markers URA3, HIS3, TRP1, and LEU2, respectively. The existing vector architecture supports plasmid propagation in *E. coli*, maintenance in yeast, and preparation of donor DNA for chromosomal integration.

Each plasmid will contain two expression units arranged in divergent orientations, with a central promoter region and terminators flanking the two coding sequences. Golden Gate assembly interfaces will be incorporated into the vector design to maintain the correct orientation and order of DNA fragments.

The eight-gene design is divided into four functional modules.

Module	Proposed promoter-gene combinations	Intended function
Precursor supply	pCCW12-tHMG1; pTDH2-ERG20 (F96C)	Increase isoprenoid precursor availability
Carotenoid biosynthesis	pTEF1-CarRP; pGK1-CarB	Introduce the β-carotene biosynthesis pathway
Lipid metabolism	pHSP26-DGA1; pTEF2-DID2	Investigate lipid metabolism and potential carotenoid storage support
Lipid and redox support	pENO2-OLE1; pADH1-tPOS5	Investigate lipid composition and intracellular redox balance

The precursor supply module is intended to improve the availability of metabolic intermediates required for carotenoid synthesis. The carotenoid module introduces the enzymes responsible for β-carotene production, while the remaining modules explore potential improvements in lipid-associated storage and metabolic support.

4.1. Bidirectional Promoter Optimization

The initial constructs will use a fixed promoter configuration to validate the assembly and integration workflow. Once the baseline system is established, selected previously characterized bidirectional promoters may be incorporated to examine the effects of expression balance on β-carotene production. A systematic characterization of new bidirectional promoters is not included in the initial scope of this project.

5. Experimental Strategy

Stage 1. In Silico Plasmid Design and Sequence Verification

The assembly workflow will first be designed and evaluated in SnapGene. Individual promoter regions, coding sequences, terminators, selection markers, and delta integration elements will be mapped onto their corresponding plasmid backbones. Type IIS recognition sites and assembly overhangs will be selected to ensure compatibility between adjacent fragments. Internal restriction sites that interfere with Golden Gate assembly will be identified and removed where necessary. For coding sequences, synonymous nucleotide substitutions will be considered to preserve the encoded protein sequence. The proposed assemblies will be simulated in SnapGene to verify fragment order, orientation, and sequence continuity.

A construct record will be prepared to document the DNA templates, primers, expected PCR products, and destination backbones. PCR amplification will be used to prepare the required inserts and vector backbones. For oligo-pool-derived fragments, amplification specificity and fragment identity will be assessed before assembly into complete genes or expression modules. The assembly will follow a modular strategy rather than combining the entire eight-gene pathway in a single reaction. Following initial validation, the workflow may be expanded to a 96-well format to support parallel construction and screening.

Stage 2. Golden Gate Assembly and Plasmid Validation

The combined DNA fragments and vector backbones will be assembled using Golden Gate assembly, followed by transformation into *E. coli* for plasmid propagation. Candidate colonies will be screened by colony PCR to identify constructs containing the expected inserts. Positive clones will undergo further verification through restriction analysis and sequencing to confirm the correct assembly of promoter-gene combinations and plasmid integrity. Verified plasmids will be used for subsequent yeast transformation and chromosomal integration experiments. If equivalent constructs generated using the original Gibson assembly workflow are available, assembly success rates and the verification effort required for each method may be compared.

Stage 3. eGFP Reporter Integration and Quantitative Screening

Before introducing the complete β-carotene pathway, the chromosomal integration workflow will be evaluated using eGFP as a reporter gene. Verified donor constructs will be prepared for transformation into *S. cerevisiae* using the laboratory's established CRISPR-Cas9-assisted delta integration system. Genomic PCR will be used to confirm the expected integration junctions. Following transformation, eGFP fluorescence will be measured to screen candidate yeast clones and identify strains with different reporter expression levels. Appropriate background controls and consistent culture conditions will be used to support comparisons. Selected clones will then undergo qPCR analysis to estimate integrated eGFP copy number, with UBC6 proposed as the reference gene. Reference assay suitability and calibration will be evaluated before quantitative comparisons.

Fluorescence intensity and integrated copy number will be analyzed separately to investigate their relationship. Although fluorescence can provide an initial indication of reporter expression, it does not directly establish the number or integrity of chromosomally integrated DNA copies. For strains carrying pathway constructs, representative gene regions and integration junctions will also be examined to assess construct integrity. The previously reported integration of up to 56 copies using DAYS provides a benchmark for evaluating high-copy integration. Comparisons will account for differences in construct size, integration design, and quantification methods.

Stage 4. β-Carotene Pathway Validation and Production Analysis

Following validation of the reporter integration workflow, the eight-gene β-carotene pathway will be assembled and introduced into *S. cerevisiae*. The resulting strains will be evaluated for successful chromosomal integration and β-carotene production. Product titres will be quantified using HPLC with suitable extraction procedures and analytical standards.

Biological replicates and appropriate reference strains will be included to assess the reproducibility of production. β-Carotene titres will be analyzed alongside biomass accumulation and cell growth to distinguish changes in metabolite production from differences in strain growth. The analysis will examine the relationships between pathway integration, transgene copy number, and metabolite production. This will help determine whether increased integration copy number is associated with improved β-carotene output. Once a baseline pathway configuration is established, selected promoter substitutions may be introduced to investigate how gene expression balance affects pathway performance.

6. Expected Outcomes and Limitations

The primary expected outcome is a Golden Gate-compatible DAYS workflow for modular assembly of multigene expression constructs in *S. cerevisiae*. The project aims to establish reliable procedures for DNA construction, plasmid verification, and quantitative evaluation of chromosomal integration.

The eGFP reporter system will provide an initial method for screening integration outcomes and selecting candidate clones for further analysis. Subsequent assembly of the β-carotene pathway will provide a functional demonstration of the platform and allow integration performance to be assessed alongside metabolite production. Several factors may influence the experimental outcome, including Golden Gate assembly compatibility, the integrity of repetitive genomic integrations, variation in reporter expression, and potential metabolic burdens associated with increased pathway copy number.

Higher transgene copy numbers may not necessarily result in increased β-carotene production, particularly if gene expression becomes unbalanced or imposes additional metabolic demands on the host. The experiments will therefore be conducted in stages, beginning with assembly validation and reporter integration before progressing to full pathway construction and production analysis. If full pathway evaluation cannot be completed within the project period, the validated Golden Gate assembly workflow and quantitative reporter-integration system will serve as the main project deliverables.

These results will provide a basis for future optimization of multigene expression, bidirectional promoter configurations, and metabolic pathway performance using the DAYS platform.